
Empirical Verification Complexity: A Research Programme for Scalable Oversight and Steering in Autonomous Laboratories

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Abstract

Verification of machine-generated science is usually attempted over the published record. We take it to belong in the autonomous laboratory, where machine-generated results are produced and where the measurement that would settle a claim can be performed, and measure one instance of it. This paper defines a verification contract, which fixes what counts as having checked a claim, and builds a benchmark on a published dispute over an autonomous laboratory’s synthesis claims: 95 targets across six groups, drawn from the original claims, an independent critique of them, and the correction that adjudicated each one. The benchmark tests whether a verifier, given only the deposited evidence the expert panels held, independently reaches the findings they published. One architecture satisfies the contract: a model proposes checks and rival hypotheses, and every verdict is re-established by a small independent, trusted kernel. This verification layer reaches 55 of 55 stage-one targets; the best unlayered condition reaches 41. No verifier reproduces the published adjudication from the evidence available when the claims were made, including the system the benchmark scores against. Two properties follow. The layer produces the data steering consumes, in that what it asks for next depends on the evidence it already holds. And oversight is scalable, in that the component that must be trusted for a refutation to count grows with the variety of the record rather than its volume. The measurements ground a programme: what a claim costs to audit is a measurable quantity, the laboratory is where it is measured, and behind it sits the larger cost the paper names alethic complexity, of turning what is true into what is known.

1 Introduction

Machine-generated science has a verification bottleneck [Cornelio et al., 2025]. Autonomous laboratories widen it, since a machine that runs experiments produces results and not only hypotheses [Abolhasani and Kumacheva, 2023, Ghareeb et al., 2025, Mitchener et al., 2025]. They also change what verification can be. Existing systems for checking machine-generated claims work over the record as published: they retrieve documents, test entailment, or re-run analyses on data that already exists [Thorne et al., 2018, Pan et al., 2023, Huang et al., 2025]. A laboratory can do what literature cannot, which is perform the measurement that would settle the question. This paper takes verification of autonomous science to belong in the laboratory, and measures one instance of it.

Where, how, and which verification happens is one question in empirical verification complexity,¹ which studies how scientific claims are made auditable and how scientific activity can be steered when claims and results become abundant. It asks how a claim can be reduced to independently checkable obligations, what observations and adjudicative work are required to discharge them, and how limited experimental capacity should be directed across testing, replication and retirement.

In November 2023 an autonomous laboratory reported synthesising 41 new inorganic compounds in seventeen days [Szymanski et al., 2023]. Within months an independent re-examination of the deposited evidence disputed much of the record [Leeman et al., 2024], and in January 2026 the original authors’ correction adjudicated 40 of the claims, confirming 36 and finding 4 inconclusive from the available X-ray analysis [Szymanski et al., 2026]. The disagreement ran compound by compound, between deposited evidence and named rival explanations. Claims, evidence, rivals and verdicts are the resolution at which this paper’s objects are defined, and both adjudicating sources published their reasoning at that granularity, which is what supplies per-claim reference labels. The deposits are public, so the record rebuilds and the grounding can be checked by the same criteria the paper applies to a verifier.

In a closed experimental loop, choosing the next experiment is a steering decision, and systems are being built to make it by reading the published literature [Ghareeb et al., 2025, Mitchener et al., 2025]. That literature is a filtered record: across every discipline surveyed, the share of published papers reporting positive results rose by more than twenty-two per cent between 1990 and 2007, this may yield “a biased, perhaps untrue, representation of what exists in nature” [Fanelli, 2012, Journal of Negative Results, 2026]. What it omits carries usable signal, a model trained on recovered notebooks of failed syntheses outperformed human intuition in selecting reaction conditions [Raccuglia et al., 2016]. An audited record, a kind that this verification layer produces, carries verdicts, the rivals that survived them, and the measurements that would separate those rivals, which is the navigable state a steering system needs.

Attention, experimental resources and judgment remain science’s scarce quantities, and their allocation is steered by institutions built for human-paced production and operating at publication cadence: refereeing, funding, replication, editorial selection and taught taste [Simon, 1971, Kitcher, 1990, Weisberg and Muldoon, 2009]. Those institutions read prose, and prose does not carry the objects an allocation decision consumes. When claims arrive faster than the institutions can read them, allocation passes by default to the generators themselves [Agrawal et al., 2024, Tao, 2026]. The question becomes which parts of the scientific landscape should be explored, challenged, replicated or left alone, and by what. The answer determines where scarce experimental capacity is spent, and in domains where experimental throughput gates progress, such as materials for energy storage and carbon capture, how quickly viable candidates are identified.

Oversight scales only when checking a claim or interpreting a result costs less than producing it, which is what makes a weaker overseer viable [Bowman et al., 2022]. For code and mathematics, checking is computing. For a claim about the world, the check needs the evidence the claim rests on. Autonomous laboratories are therefore what makes verification urgent and what makes it possible: they generate results faster than those results can be judged, and they are also where the measurement that would settle one can be performed.

Autonomous laboratories make this a safety problem as well. Their outputs become inputs to subsequent physical actions without waiting for publication or review, so an erroneous interpretation can determine what is synthesised next. A verification layer supplies what a safety policy needs and does not itself provide: a point at which a proposed action can be stopped, and the evidence trail on which stopping it can be justified [Zhang et al., 2026, Leong et al., 2025]. That matters most in areas such as biosecurity [Urbina et al., 2022]. The interfaces that make autonomous execution possible, including plans, instrument commands, logs and provenance, are what would make such a point enforceable [Moreau and Missier, 2013]; whether they suffice is open.

A verification layer produces what a steering decision consumes: a verdict, the rivals that survived it, and the measurement that would separate them. It does not make that decision, in the way a proof assistant checks arguments without determining which theorems mathematics should pursue [de Moura and Ullrich, 2021]. The open problem is to design and evaluate verification-conditioned

¹Three neighbouring terms are occupied: *verification complexity* in proof theory [Arana and Stafford, 2023], *empirical complexity* in computational epistemology [Kelly, 1996, 2007], and *verification complexity* in systems engineering [Jung and Salado, 2024]. The compound names a different object.

steering policies: policies that decide which experiments to perform, which claims to challenge or replicate, and which directions to defer, while keeping those decisions auditable.

The paper contributes three things. First, a verification contract, which fixes what counts as having checked a claim, and a benchmark built on it, where an outside submission is scored by the same scorer as the runs reported here. The benchmark supplies 55 targets from the evidence deposited with the original claims and 40 more from an analysis performed after the dispute.

Second, one architecture, a verification layer that satisfies the contract, and what it measures. A model proposes and a small independent component decides. Across four models and four conditions the layered architecture reaches all 55 stage-one targets, where the best unlayered condition reaches 41. What it asks for next depends on the evidence it already holds. The component that must be trusted for a refutation to count does not grow with the size of the record.

Third, a finding about where verification has to happen. Working from the evidence available when the claims were made, no verifier reproduces the published adjudication, including the system the benchmark scores against. Granted the analysis the layer asked for, it reaches that adjudication on all 40 compounds.

Those measurements are one record. Three conjectures generalise them: that a trusted kernel for empirical claims can be built at all, that the cost of oversight tracks the variety of a record and not its volume wherever the record comes from, and that verification inside the laboratory reaches what verification over the published literature cannot. Section 5 states each with what would refute it.

2 The benchmark

The benchmark asks whether a verifier, given only the deposited evidence an expert panel held, independently reaches the findings that panel published.

The A-lab episode is a claim-level dispute in which both sides published per-claim reasoning, so the record rebuilds end to end: the original synthesis claims [Szymanski et al., 2023], the published critique of them [Leeman et al., 2024], and the Correction that adjudicated each claim individually [Szymanski et al., 2026].

2.1 The verification contract

A claim is one assertion the source makes about one compound, together with the evidence deposited alongside it. A rival is an alternative account of that same evidence, which a verifier deciding for the claim would have to rule out. A verifier declares which rivals it considered; the contract does not fix how they are chosen.

A verifier has five verdict/outcome states: REFUTED, UNDECIDABLE-AT-BUDGET, BOUND-INCONCLUSIVE, CONSISTENT and SET-INADEQUATE. They don't affirm a claim as a refutation cites evidence a third party can re-examine, whereas an affirmation doesn't. This grammar inherits the falsifiability asymmetry [Popper, 1959].

REFUTED records that a check found evidence that contradict the claim. CONSISTENT records that nothing contradicted the claim, and that the measurement available was enough to separate it from the rivals. SET-INADEQUATE records that neither the claim or its rivals together explain the analysis, and do not account for what was measured. UNDECIDABLE-AT-BUDGET records that the measurement available could not have separated the claim from the rivals. BOUND-INCONCLUSIVE records that neither of those last two could be shown.

Every verdict declares the conditions under which it holds: the rival class it was decided against, the instrument model assumed, the evidence stage held, the measurement budget, and the error rate. The evidence stage is the benchmark's, since it controls what is released; the other four are the submission's, and the budget is declared per instance rather than fixed across the population.

A refutation counts only when an independent checker re-establishes it [Thorne et al., 2018]. Validity is an established contradiction between the deposited evidence and the claim record. The requirement is read from the claim record rather than from the witness text, because that text is written by the system being judged. The checker uses the standard library only and shares no code with any verifier.

An outside verifier needs none of this project’s layer, models, primitives or code. It submits one JSONL file, and the same scorer and rubric that produced the results in Section 4 produce its score. The benchmark distribution² gives the submission schema and the classes of evidence a submission reports on.

2.2 The targets

The conditions for a target is that an expert source published it, and a model-written check could reach it from the claim and the deposited evidence (Figure 1). The evaluation population is the 40 claims the Correction adjudicated. **K1-K5** are the critic’s results, and **K6** is the correction’s. A deterministic harness re-derives every target from the deposit alone. The exclusions, the populations, and the harness are in the benchmark distribution.

	SOURCE	EXPERT DERIVATION	TARGETS	STAGE
K1 structure mismatch	automated structure files	critique Table I, the E2 flag	7	one
K2 symmetry	automated structure files	critique Table III, indexed_sym	38	one
K3 novelty	reference databases	critique Table I, the E4 flag	2	one
K4 lattice parameter	automated structure files	critique Table II	5	one
K5 mixed sites	automated structure files	critique section IV prose	3	one
K6 baseline decision	manual refinement columns	40 decisions, 8 flagged	40	two

55 stage-one targets, 40 stage-two

Figure 1: The ceiling. Each row reads left to right: the finding, its source in the record, the expert derivation, the number of targets surviving exclusion, and the evidence stage.

Evidence stages Stage one is the evidence that existed when the claims were made. Stage two is an expert re-refinement that exists because the dispute happened, and it supplies a re-refinement against an expanded candidate phase set. A verifier that abstains and does not reach CONSISTENT, as well as also requesting the analysis type stage two holds receives it. Note that stage-two evidence is contaminated, as it has been produced by people who knew the outcome, and scored separately.

K6 is the only group scored at stage two. It is the pass/flag decision of CARTOGRAPH [Shah and Kartik, 2026], reproduced from that paper, which was recovered from the Correction. Its inputs are manual refinement columns, so it could not have been computed at stage one.

2.3 Scoring

Scoring is per target. Five outcomes are recorded. RECOVERED: the target was reached and matched. MISSED: a check reached it and gave a different answer. UNVALIDATED: a refutation was made and the scorer could not read it. NOT-OBSERVED: a check ran and recorded nothing for that quantity. NOT-ATTEMPTED: no check that could reach the target was written.

2.4 Controls

No expert label, flag, verdict or finding appears in any input, and each compound is submitted once.

Contamination. Two directions are constructed from the evaluation set with the compound name preserved in both. H1 removes a fault from a compound the critique flagged; B1 injects one into a compound nobody flagged. The directions predict opposite behaviour, which is what makes this a control for memorisation: a verifier that works answers H1 cannot_verify and B1 refuted, while one recalling the literature does the reverse.

²<https://github.com/ArdenTSH/verification-layer-benchmark>.

Mutants. A second control set is wrong by construction, with the truth recorded outside the claim by a committed generator, across six families spanning four orders of magnitude of difficulty. Detecting a planted fault and reproducing an expert finding are different questions, and the mutants are not reproduction targets.

3 Aletheia: the verification layer

Aletheia is the instance of the architecture measured here: one primitive library, one admission library, one forward-model library, one pricing library and one checker, filled with what a single crystallographic dispute required. The architecture is the separation of proposing from deciding. Aletheia is one filling of it. The results in Section 4 are Aletheia’s; the conjectures in Section 5 are the architecture’s.

3.1 Design

The layer is built on the asymmetry of scalable oversight, between a system and the weaker overseer that checks it [Bowman et al., 2022]: an overseer can be weaker than what it oversees when checking is cheaper than producing. So the layer separates proposing from deciding. A model writes checks and proposes rivals, and it is untrusted until a program sharing no code with it re-establishes the result.

Proof assistants have a similar shape: a kernel small enough to audit, proof terms that are re-checkable objects, and search procedures that may be arbitrary because nothing they produce is trusted until the kernel accepts it [Barendregt and Geuvers, 2001]. Certifying algorithms and proof-carrying code are the same move outside logic [McConnell et al., 2011, Necula, 1997].

A proof kernel in maths cannot be fooled as mathematics only needs to be internally consistent. An empirical kernel executes under assumptions it cannot verify. So constructing trusted kernels for empirical science is an open problem. Such a kernel would have to reach the experiment, where the only trusted kernel is nature itself. Since the benchmark is retrospective, it is not done here. Autonomous laboratories are what would put an experiment within a verifier’s reach, it is both the motivation for needing scalable oversight, as well as the only place that can substantiate it. Section 5 states the problem.

Rivals are designed from the framework of multiple working hypotheses, where a claim is supported to the extent that alternative hypotheses have been excluded [Chamberlin, 1890, Platt, 1964].

Abstention is the default output, since unfalsified is the default condition of a claim [Chow, 1970, El-Yaniv and Wiener, 2010], and an abstention carries the request for evidence on which the second evidence stage is conditioned.

3.2 The chain

The layer takes one claim, and returns a vector, a verdict, the conditions it holds under, and the residual, which is the rivals still standing together with the observation that would separate them.

Decomposition and intake. A model decomposes a single claim into several child claims, each asserting one thing about one family of evidence (here as the phase present, the cation ordering, the weight fractions, or novelty against the reference databases). This is done such that an assertion is decomposed into units checked independently from established text [Wei et al., 2024]; here the library reads a physical deposit. Each unit becomes a record carrying an identifier, a pointer to its parent, and the evidence it rests on, resolved to specific deposited artifacts and hashed [Kuhn and Dumontier, 2014]. Recording which artifacts a claim rests on, so that a conclusion can be traced back to them, is standardised as a provenance model [Moreau and Missier, 2013] and implemented for autonomous research as a chain of evidence [Meng et al., 2026]. How finely a claim splits varies several-fold across models, and Section 4 reports that spread.

Elicitation. A model writes one program per child claim. Each program runs once. Agents already generate re-runnable hypothesis tests [Miao et al., 2026] and design sequential falsification experiments under formal error control [Huang et al., 2025]. This layer seals the elicitation, where

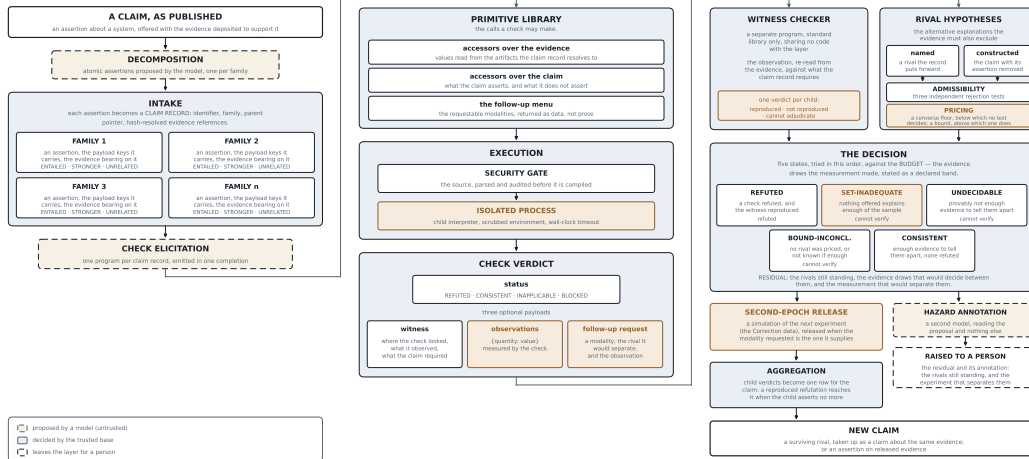


Figure 2: The chain, from the claim. Read down each column and then across: vertical position is sequence and horizontal position is alternatives.

a program reads the deposit only through a fixed set of functions, the primitive library of Figure 2, which bounds what any check can look at and returns measurements. Programs are stored and reused, so every later stage can be changed and re-run without another model call; a program is rewritten only when the set of functions changes.

Execution. Anything outside the primitive library is refused before the program runs. Across 808 executed checks the gate refused sixteen programs, each calling a function that does not exist, and the refused names are a record of what the vocabulary lacks and a reviewer can read them as candidates for the next version. The library has grown this way: three functions added at one point made three target groups reachable that no check could previously ask about. A program then separately runs the check, without credentials and under a time limit.

Check Verdict. A check returns one of three statuses: refuted, consistent and inapplicable, with a payload. A refutation carries a witness, which records where the check looked, what it found there, and what the claim required instead; that is the object the independent checker re-establishes. Any check may also record observations, which are measured values with no refutations attached. This is the common element across operational evidence graphs [Ren et al., 2026, Meng et al., 2026].

A check may also name the follow-up measurement that would settle the question, drawn from a fixed menu. Choosing the next observation to discriminate hypotheses is owned by active and sequential testing and by Bayesian experimental design [Chernoff, 1959, Naghshvar and Javidi, 2013, Chaloner and Verdinelli, 1995], and previous work in steering, and stopping in an autonomous laboratory includes [Shah and Kartik, 2026, Kusne and McDannald, 2026].

Rival hypotheses and AI Scientists. A rival is an alternative account of the same evidence. The record names some, as provided by the original papers here; where it names none, the layer constructs one by disordering the claim’s own structure.

More importantly, the slot is also where a new hypothesis enters. Any hypothesis worth proposing contradicts something, whether an account the record holds or the absence of one, so it enters as an alternative to what the record currently says. That makes the rival slot an interface for hypothesis generation, which is where autonomous-science systems concentrate their effort [Ghareeb et al., 2025, Mitchener et al., 2025, Lu et al., 2024, Google, 2025]: a proposal may come from any system, is admitted and priced like any other, and is excluded from the verdict. High-throughput AI generated science can therefore be used without being trusted. Generating those alternatives from the data is an active research area, and automated multiple-hypothesis phase identification is the nearest attempt [Fei et al., 2025] from the group whose claims this benchmark rebuilds [Szymanski et al., 2023]; the layer accepts its output as a rival source and has run it.

Admission and pricing. A rival is admitted when it passes admissibility tests, the trusted base’s equivalent of the primitive library. The tests check form rather than scientific plausibility, so a proposer can inflate a claim’s apparent difficulty with a well-formed implausible rival. Each admitted rival is then priced against the claim. Two quantities travel with the pair: the achievable bound says a measurement of a given size is enough to separate them, and the floor says none below it can [Chernoff, 1952, Le Cam, 1973, Tsybakov, 2009]. Choosing an experiment by how much it would discriminate is established [Chaloner and Verdinelli, 1995], and relating how much measurement is taken to what it can resolve is long-standing in diffraction [Hegerl and Hoppe, 1976].

The instance decision. Each child claim receives one of the five states of Section 2.1 by an ordered rule, first match wins: a witnessed refutation, then set-inadequacy, then the three pricing regions, which are the floor above what the measurement provides, the achievable bound within it, and neither. Set-inadequacy is asked before the pricing regions because if the offered hypotheses do not account for the sample, pricing claim against rival compares two explanations neither of which explains what was measured. This state is computed by the trusted base.

Aggregation. Aggregation collects the verdicts on every claim and every compound into one place. Each verdict carries a residual stating what remains open on that claim and what would settle it, and those are collected with them. A choice of next experiment reads both. The layer currently writes this as one row per claim.

How the space should be represented is open. Typed evidence graphs are one attempt [Ren et al., 2026, Meng et al., 2026], and the claim-graph lineage of nanopublications, micropublications and provenance vocabularies supplies the substrate without a kernel that decides [Groth et al., 2010, Kuhn and Dumontier, 2014, Clark et al., 2014, Moreau and Missier, 2013]. How to act on it is a second open problem; retrieval-augmented generation [Lewis et al., 2020] and reinforcement learning with verifiable rewards [Shao et al., 2024, Lambert et al., 2024] are the obvious first attempts. Section 5 returns to both.

Residual. The residual leaves the layer naming a measurement or experiment that the lab should perform next. Whether that measurement is hazardous is not a judgement this layer currently makes: hazard is a domain judgement, and the trusted base holds only what is small enough to read. The layer annotates the residual and does not gate it. What stops a proposal is a separate authority acting on the record the layer produces, and the composition of the two is open (Section 5).

3.3 The trusted base

Five of the layer’s components are libraries. The primitive library holds the functions a check may call to read evidence. The admission library holds the tests that decide whether a proposed rival hypothesis is valid. The forward-model library holds the methods that turn a hypothesis into a predicted observation. The pricing library holds the methods that say how much observation separates two hypotheses. The checker holds one predicate per class of evidence. The decision rule and the aggregation rule are fixed. Aletheia’s trusted base is 3,757 lines.

A record drawing on other instruments or other kinds of claim adds entries to these libraries. This is by design, these libraries should continue to be expanded, but expanded in versions, as well as upheld openly, sourced communally, and remain human-legible. How trustable these libraries are, as well as how they scale across domains remain as open problems, see Section 5.

4 Results

4.1 Conditions

Four models ran through Aletheia on the same 40 compounds using stage-one evidence. Three unlayered conditions ran on one of those models: reading the deposited file with no execution; writing and running one program without seeing the file; and writing and running one program with the file supplied. Six mutant compounds ran through all seven configurations as a contamination control.

A second round ran on one model with stage-two evidence readable by eliciting follow-up checks. It is a different population from the four-model comparison.

Table 1: Target recovery. The two blocks share the group spine, and the layered cell appears in both. Group denominators are not pooled. K6 is scored at stage two and is reported in Table 4; it has not been elicited for the other three models. The one target the file-seen condition did not attempt is counted against the benchmark’s denominator. Every cell in Table 1 is one model and one run, with no intervals.

Block A — four models, all through the layer					
group	targets	opus5	sonnet5	gpt56	gpt56-terra
K1 file differs	7	7	5	2	3
K2 determination = claim	35	35	35	35	35
K2 requires the file	3	3	3	3	3
K3 already known	2	2	2	2	2
K4 lattice parameter	5	5	0	1	1
K5 prose	3	3	3	3	3
Block B — one model (opus5), four conditions					
group	targets	reading only	one program file unseen	one program file seen	the layer
K1 file differs	7	4	5	2	7
K2 determination = claim	35	31	20	34	35
K2 requires the file	3	2	3	2	3
K3 already known	2	0	0	0	2
K4 lattice parameter	5	1	0	0	5
K5 prose	3	3	3	2	3

4.2 Target recovery

The layered opus5 run reaches the ceiling on all six groups, exceeding every unlayered condition on K1, on K2’s echo half, on K3 and on K4, and tying on the remaining two (Table 1). Among the four models, K1 and K4 separate models, at 7/5/2/3 and 5/0/1/1; K2, K3 and K5 return identical scores for every arm, each at its ceiling.

Across conditions the largest gap is K2’s echo half, at 31/20/34/35. Most of that half is earnable by reporting the claimed space group, so the informative comparison is the blind condition’s 20 against the file-seen condition’s 34: writing a program against a file it cannot inspect costs more than not writing one at all.

4.3 Contamination control

Six of the seven configurations are correct on both healed compounds; the one miss is the condition that never saw the file, which refuted on the deposited P 1 header (Table 2). The remaining failures fall in the detection direction, where injected disorder on three of four compounds goes unreported by two models.

Table 2: Contamination control. Seven configurations over six constructed compounds; completion and accuracy are reported separately. H1 is the memorisation-sensitive direction. B1 is coarser than the benchmark, at eight to twelve shared cation sites, so a high B1 score does not indicate that the benchmark is easy.

configuration	checks completed	H1 correct	B1 correct
reading only	6 of 6	2 of 2	4 of 4
one program, file unseen	6 of 6	1 of 2	4 of 4
one program, file seen	6 of 6	2 of 2	4 of 4
layer, opus5	6 of 6	2 of 2	4 of 4
layer, sonnet5	6 of 6	2 of 2	2 of 4
layer, gpt56	6 of 6	2 of 2	1 of 4
layer, gpt56-terra	6 of 6	2 of 2	1 of 4

Table 3: What follow-up request the layer asks for, by what it already holds. Columns differ in whether the re-refinement had already been released to that child.

modality requested	lacks it	holds it
re-refinement, expanded phase set	75	11
higher-resolution XRD	3	54
elemental analysis	1	14
electron diffraction	0	4
neutron powder	0	3
longer counting	0	1
total	79	87

Table 4: The second evidence stage. Same claims, same model, same population; the two rounds differ in whether the requested analysis was available when the checks were written.

	round one	round two
checks written	blind to stage two	with stage two granted
agreed where the source asserts	4 of 7	8 of 8
agreed where the source does not	27 of 31	32 of 32
overall	31 of 40	40 of 40
recorded no usable observation	2	0

4.4 Layer output

Over 808 child instances pooled across the four models, four of the five instance states occur: BOUND-INCONCLUSIVE 382, CONSISTENT 203, REFUTED 185 and SET-INADEQUATE 38. UNDECIDABLE-AT-BUDGET is the only state the floor certifies, and it occurs zero times. The floor carries a calibration constant, and that constant has not been calibrated against this instrument. Because the benchmark is retrospective, no measurement can be commissioned to calibrate it, so the constant is set where it cannot fire spuriously; the price of that setting is that it does not fire at all. The state stays in the contract, since a submission working prospectively can certify it, and calibrating a floor against a named instrument is one of the open problems of Section 5.

Pricing runs on 214 of the 808, on all 203 CONSISTENT instances and on 11 of the 38 SET-INADEQUATE, approximately a quarter of total.

Four of the forty claims carry a rival that a person proposed; for the rest the layer constructs an alternative by disordering the claim’s own structure. The tool-proposed path was exercised, with an external phase-identification system: its proposals were admitted, priced and reported.

4.5 Steering

When the layer chooses an abstaining verdict, it names the follow up request that would settle the question, committed in the same completion as the check. What it names depends on what it already has. Children that had not been given the re-refinement asked for it in 75 of 79 cases; children that had been given it asked for it in 11 of 87, and asked instead for higher-resolution diffraction in 54 (Table 3).

To simulate steering, stage-two evidence was hidden and released only on request.

K6 is scored on stage-two evidence. The layer worked on stage-one evidence, abstained, named the analysis that would settle the question, was granted it, and was scored against the external comparator’s decision. Round one scores 31 of 40 and round two 40 of 40, agreeing on all 8 compounds the source flags and all 32 it does not (Table 4).

4.6 Scalable Oversight

In Aletheia, what must be trusted for a refutation to count is 1,015 lines using the standard library only, sharing no code with the layer or with the models that wrote the checks. It grows with the number of evidence classes rather than the number of claims: one predicate per class, and adding

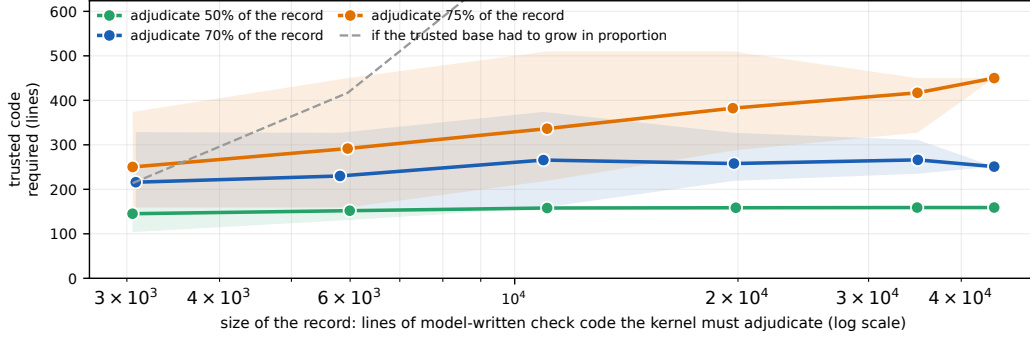


Figure 3: The trusted base against the record it adjudicates. For each record size, the smallest set of checker predicates that adjudicates the stated share, averaged over repeated subsets of the record, with the shaded band covering the tenth to ninetyth percentile. The dashed line is proportional growth.

compounds adds none. Figure 3 shows the consequence: the trusted code needed to adjudicate a fixed share of the record stays flat as the record grows.

Once compiled, a check is stored, so changing the decision rule and re-adjudicating the whole record costs no model calls.

4.7 Limitations

The contract specifies five verdict states; this record exercises four. Pricing runs on 214 of 808 instances and the floor certifies none of them, so the measurement-budget half of the contract is demonstrated as a mechanism and not evaluated as a method. What the layer is shown to do here is decompose, seal, re-establish, and abstain with a request. What it is not shown to do is price a claim against a rival well enough for the price to decide anything.

Thirty-five of the 55 stage-one targets are the echo half of K2, most of which is earnable by reporting the claimed space group. The comparison that carries weight is the 20 remaining targets, where the layer reaches 20 and the best unlayered condition reaches 11.

The evidence is one episode and one measurement class, with limited data. So several of the comparisons above rest on populations too small to reach statistical significance, and are reported as consistent with a reading rather than as establishing one.

5 Outlook and conclusions

Hypothesis generation. The proposer sits outside the trusted base and what it proposes is priced. Pricing gives a proposer an objective, since for any admissible rival the trusted base returns the measurement that would separate it from the claim, and proposing then becomes a search for the admissible alternative that costs the most to exclude. That objective is a number, so a proposer can be trained against it. The open problem is an admission gate that holds against models that game it [Anil et al., 2021, Skalse et al., 2022].

Representation. Aggregating into one row per claim cannot express that a verdict on one claim obliges a re-check of another. Which verdicts carry that obligation and how far it propagates is what a representation has to make computable, and it is the open question the typed-graph programmes are closest to answering [Ma, 2026].

Steering. Several claims stand open at once, each residual names a measurement, and a laboratory has one instrument. Choosing among residuals is an allocation problem this paper leaves untouched, and the measurement budget should follow what a claim is worth rather than a fixed count per instance. The record a steering policy would train on is also different in kind from the published literature, because it keeps what publication filters out: refutations, abstentions, and the rivals that survived. Failed syntheses have already outperformed human intuition as training data once [Raccuglia et al.,

2016]; whether a policy trained on an audited record steers better than one trained on the literature is a measurable question.

The libraries. This instance holds a primitive library, an admission library, a forward-model library, a pricing library and a checker, for one class of measurement. A record drawing on other instruments or other kinds of claim is adjudicated by adding entries, so the libraries are where the architecture grows, and how they grow determines what the architecture is worth. They should be open source, versioned, and kept by the communities whose measurements they encode, the way mathematics keeps a shared library of formal proofs [The mathlib Community, 2020]: a verdict cites the library version it was decided under, a library change names the verdicts it reopens, and what the trusted base contains is public in the same sense the claims it judges are. Each discipline already holds candidate entries. Crystallography has a long-established answer for comparing models that differ in the number of free parameters [Hamilton, 1965], and a pricing library should reach for it where it applies. What is open is transfer: whether libraries of this shape carry from one domain to another, and what a pricing library holding several statistical methods selects among them on.

Safety. An autonomous laboratory acts on its own results without waiting for review, so an interpretation determines what is synthesised next. The layer supplies the point at which a proposed action can be stopped and the evidence trail on which stopping it can be justified. Hazard models, containment and instrument state sit outside it [Zhang et al., 2026, Leong et al., 2025, Tang et al., 2025]. How an evidence-admission contract composes with an action-permission contract is open, and that composition decides whether an audited record can govern what a laboratory does next.

The kernel. Every artifact the layer reads stands in for a measurement made under conditions the layer cannot check. Calibration, specimen identity, instrument state and specimen history enter as declared assumptions. A kernel for empirical claims would have to reach the experiment, and an autonomous laboratory is what would put an experiment within a verifier’s reach. That is the first conjecture, that such a kernel can be built at all. An argument that no empirical primitive can authenticate itself would refute it.

Scalability. The checker holds one predicate per class of evidence, so the trusted code needed to adjudicate a fixed share of the record stays flat as the record grows (Section 4.6). The second conjecture is that this holds wherever the record comes from: the cost of oversight tracks the variety of a record and not its volume. A domain in which the number of kinds of evidence grows with the number of claims would refute it.

The laboratory advantage. The third conjecture is that verification inside the laboratory reaches what verification over the published record cannot. The measurement behind it: at stage one no verifier reproduces the published adjudication, and granted the analysis it asked for the layer reaches it on all forty compounds (Table 4). It has a formal ancestor, since for causal graphs verifying a proposed answer is proven to need fewer interventions than finding one [Choo et al., 2022], and interventions are the resource an autonomous laboratory spends. If the conjecture holds, verifiers belong where the discriminating measurement can be run, and a verification layer becomes standard laboratory equipment rather than a stage of publication. Two findings would refute it: a domain in which the discriminating experiment costs more than the claim is worth, and a domain in which the published record suffices at scale.

Alethic complexity. Empirical verificational complexity is the cost of making a claim auditable: the observations, the adjudicative work, and the trusted code a verdict on it requires. That is what this paper measures. Auditability serves a wider question, which is what it costs to turn what is true into what is known, and *alethic complexity* names that cost. The adjective carries a philosophical sense concerning truth as against justification, and it is used here for the wider question.

This paper defines a verification contract, builds a benchmark on an adjudicated dispute over an autonomous laboratory’s synthesis claims, and measures one architecture against it. The layered verifier reaches 55 of 55 stage-one targets where the best unlayered condition reaches 41, what it asks for next depends on the evidence it already holds, and the component that must be trusted for a refutation to count does not grow with the record. A layer of this kind, running inside the laboratory

at the time of deposit, would have surfaced what a separate critique was later required to establish, and would have declined the questions that needed an experiment nobody had yet performed.

What is proposed here is an architecture rather than any one system. A model proposes, a small independent component decides, and everything that component must trust is a library indexed by kind and small enough to read. This specific benchmark fills those libraries with what one crystallographic dispute required. Aletheia is one system; what the architecture is worth depends on what fills the libraries next.

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The deposited evidence this benchmark rebuilds was made public by the authors of the original study and its critique, and the external comparator’s decisions are reproduced from its published definition.

The benchmark, scorer, and harness are at <https://github.com/ArdenTSH/verification-layer-benchmark>. The layer, libraries, and kernels are at https://github.com/ArdenTSH/Aletheia_v0

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